

CONCENTRATION, NOT TARGETING: A PRE-REGISTERED NULL FOR UNCERTAINTY-GUIDED SYNTHETIC DATA IN CAMOUFLAGED OBJECT DETECTION

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ABSTRACT

Camouflaged object detection asks a model to outline objects that blend into their background. Labels are expensive — tracing a boundary you can barely see is slow — so synthetic camouflaged images are a common partial remedy. But which images should you generate?

We built and tested an answer: cluster the unlabelled real images the model will face, find the clusters it is least sure about, generate images aimed there, retrain, repeat. The ingredients are standard — active learning, mean-teacher disagreement, clustering — but the loop is ours.

It does not help. Across 36 controlled training runs, aiming the budget by uncertainty does not beat spending it at random; our design could detect an effect the size of the one reported for this approach, but not one several times smaller, resolving 39% to 45% of it. More interesting is why the idea looks like it works. Targeted selection does pick a visibly different training set from a random one — but so does any rule that concentrates the budget on a few clusters, including one that keeps our allocation’s shape while destroying the uncertainty signal in it. The separation comes from concentration, not targeting; the discriminating control is a same-shape shuffle.

Independently: 41 of CHAMELEON’s 76 images (53.9%) are re-encoded copies of COD10K and CAMO training images. Exact hashing, the standard check, reports zero overlap because the copies were re-saved in a different format. We claim non-independence rather than inflation, and release a detector and a clean protocol. A control that returns zero is informative only if it could have returned something else.

1 INTRODUCTION

Camouflaged object detection asks for a pixel-accurate outline of an object that is hard to see on purpose: an animal whose colouring matches the leaf litter it sits on, an insect shaped like a twig. Its training masks are unusually expensive, because an annotator must decide where a boundary lies when the point of the object is that its boundary is invisible. Labelled data is correspondingly scarce, and one response is to manufacture more: image generators can place a real foreground object into a freshly generated background and produce a plausible camouflaged scene. But generation is not free, and every generated image has to be trained on. So if you can make a thousand images, which thousand?

Our answer. Let the model choose: train it, find the unlabelled real images it is least sure about, aim the generation there, retrain, repeat, so each round of new data lands where the model is weakest. We built such a loop and tested it. Every ingredient is standard — uncertainty sampling from active learning, a mean-teacher disagreement score as the uncertainty estimate, clustering over self-supervised features so the budget spreads rather than piling onto near-duplicates. The composition is ours, and it does not work.

What we found, and how far the measurement reaches. Across 24 training runs — two architectures, four conditions, three seeds — a targeted allocation does not beat a random one: the decisive gap is WITHIN NOISE on both architectures and both endpoints. The qualification is that this campaign resolved coarsely: its threshold was 126% of the improvement the published approach reports, so it could not have detected that improvement had it been present. A second campaign of 12 runs tightens the threshold by $3.24\times$, and under *that* bar the decisive gap sits at $0.91\times$ threshold — still within noise, marginally so. Both are therefore *no effect resolvable at this sensitivity*, never a demonstration of absence. A mean-teacher positive control shows the instrument is not blind, detecting a real effect of $+0.0197$ at $3.3\times$ its bar — though not, as Section 5 states, an effect the size of the nulls.

Why it looks like it works, and a second finding. Targeted selection really does produce a training set measurably different from a random one — but so does any rule that concentrates the budget on a few clusters, including rules that keep our allocation’s exact shape while destroying the uncertainty signal inside it. What looked like the effect of aiming was the effect of concentrating, so the discriminating control is not random selection but a same-shape shuffle. Separately: the CHAMELEON benchmark is more than half training data, and the standard check returns clean. Some of what follows is specific to the loop we built and some travels; we say which for each. One pattern runs through both — a control that returns zero is informative only if it could have returned something else.

Contributions. The first is independent of our method, the next three travel beyond our setup, and the last is a property of the loop we built.

- 41 of CHAMELEON’s 76 images (53.9%) are re-encoded COD10K-train and CAMO training images, while exact hashing returns 0. Verified against an author-sourced release and calibrated by a negative control that finds none where none should be. We claim non-independence rather than inflation, and identify a published protocol whose unlabelled target pool holds 40 of the 76 test images (Section 7).
- The target distribution has no cluster structure to allocate over — best silhouette 0.1600, no better in three embedding spaces (Section 6.2).
- An uncertainty signal can contribute nothing beyond the concentration it induces: $+0.0073$ of a Cohen’s d against its own shuffle, and with concentration held *exactly* fixed, real, destroyed and reversed targeting are indistinguishable in 12/12 cells (Section 6.1).
- Uncertainty predicts pixel error better than structural error in six independent signal-architecture tests, surviving an area control that clears its floor by no margin on one of two covariates (Section 6.3).
- Why this loop could not have shown a gain: a pinned optimisation budget, a foreground supply that is a bijection onto 4447 copied objects, and a generated distribution narrower than its own input; we also *withdraw* one of our own candidate causes (Section 6.4). None licenses the conclusion that closed-loop generation cannot work.

2 RELATED WORK

Camouflaged object detection and its benchmarks. The task asks for pixel-accurate segmentation of objects whose colour and texture match their surroundings [CITE: COD task definition / survey], on a small fixed set of benchmarks — COD10K [CITE: COD10K benchmark], CAMO [CITE: CAMO benchmark], NC4K [CITE: NC4K benchmark] and CHAMELEON [CITE: CHAMELEON benchmark] — with search-and-identification architectures [CITE: SINet], [CITE: SINet-v2], [CITE: SegMaR] and structure-aware measures [CITE: S-measure], [CITE: E-measure], [CITE: weighted F-measure]. That those four are the standard reporting set is what makes Section 7 matter beyond any single method.

Synthetic data and generative augmentation. Diffusion-generated training data is a standard response to label scarcity in dense prediction — tasks that label every pixel — [CITE: diffusion-generated training data for segmentation], [CITE: latent diffusion], [CITE: DDIM], with compositing augmentation its cheaper predecessor [CITE: copy-paste augmentation]. For camouflage, a

synthetic-to-real line [CITE: synthetic-to-real domain adaptation] regenerates a background around a real foreground [CITE: LAKE-RED] and couples that generator to a self-training loop [CITE: S2R-COD]. Both are prior work, used as released; what we add is deciding *which* images the generator should make. Other camouflage generators exist [CITE: other camouflage image generation] and we do not test them, so our null is about this one.

Uncertainty-guided acquisition. Every ingredient we compose is standard, which is why the idea was worth testing. Acquiring data where a model is least certain is long-established [CITE: uncertainty sampling / active learning survey], with teacher disagreement [CITE: mean teacher], ensembles [CITE: deep ensembles] and predictive entropy [CITE: predictive entropy / MC-dropout] as estimators, and consistency training [CITE: consistency regularisation / FixMatch] and pseudo-labelling [CITE: pseudo-labelling] as the usual way to use unlabelled data. In the batch setting the choice is also spread over a representation, by clustering or core-set selection [CITE: core-set / diversity-based acquisition], here over self-supervised features [CITE: DINOv2], [CITE: CLIP]. Uncertainty and diversity are known to be entangled there [CITE: batch active learning, uncertainty-diversity trade-off]; Section 6.1 measures how much benefit survives when the uncertainty is destroyed and only the diversity remains.

Benchmark integrity and negative results. Train-test contamination has repeatedly been found in vision benchmarks [CITE: train-test contamination in vision benchmarks], [CITE: CIFAR/ImageNet duplicate analysis], and is a first-order concern for large models [CITE: benchmark contamination in LLM evaluation]; what we add is that exact hashing certifies a contaminated set as clean. We follow arguments for committing analyses in advance [CITE: pre-registration in machine learning] and for publishing disconfirming results [CITE: negative results / reproducibility in ML]. That a learned probe separates two encodings of identical content joins criticism of learned distributional distances [CITE: critiques of learned-probe distributional distance], [CITE: FID / representation-metric critiques].

3 THE LOOP WE BUILT

The idea is easy to picture. Train a model on the labelled images you have; look at the unlabelled real images it will be judged on and find the ones it handles least confidently; ask a generator for new images resembling those, add them, and train again. What follows is our instantiation, specified exactly, because a null is only as informative as the thing it tested.

3.1 SETTING AND LOOP

The setting is source-to-real domain adaptation: we train on a *source pool* of 4447 labelled image and mask pairs and want the model to work on a *target pool* of 4040 unlabelled real images, with verdicts read from two held-out *endpoints*, COD10K-test primary and NC4K secondary. The target pool is not one distribution — it mixes 3040 COD10K images with 1000 from CAMO, separable by a held-out probe at up to 0.9225 AUC — so a method partitioning it is partitioning a mixture (Section 6.2).

Training runs in two rounds. In round one a student trains on the source pool while a teacher — a slowly updated running average of the student’s weights — gives a second opinion on the unlabelled images that the student must agree with. Between rounds, target images whose student-teacher edge disagreement falls below a fixed fraction of the set average, meaning the two already agree, are appended with the teacher’s prediction as their label. Round two rebuilds both models from ImageNet initialisation and trains on the enlarged pool; its best teacher checkpoint is evaluated. Two properties matter and neither is visible from “closed loop”: the round count is a fixed constant, not a convergence criterion, and the generator is never invoked from training but renders its pool once, offline, the only returning edge carrying pseudo-labels rather than images. It composites rather than synthesises (Section 6.4).

3.2 UNCERTAINTY-GUIDED ALLOCATION

Our allocation spends a fixed generation budget where the model is weakest, in four stages.

(i) **Group the target images**, so “weak” can name a region of the data rather than one image: k -means into $k = 75$ clusters over L2-normalised self-supervised embeddings, fitted once and never refitted. (ii) **Score each image for uncertainty**. We need a number, computable without labels, saying how unsure the model is: how far the student and its running average disagree. The consistency loss already reduces that quantity, which is what makes it available on unlabelled data:

$$ES(x) = a \|\nabla\sigma(s(x)) - \nabla\sigma(t(x))\|_1 + b \text{BCE}(\sigma(s(x)), \sigma(t(x))), \quad (1)$$

with student and teacher logits s, t , sigmoid σ , Sobel gradient magnitude ∇ , and configured weights $a = 0.9, b = 0.3$; a cluster’s score es_j is the mean over its members. (iii) **Convert scores into a budget split**. Budget B is spread by a temperature softmax over es_j with $T = \alpha \cdot \text{sd}(es)$, at $\alpha = 1.0$ and $B = 1000$; the temperature sets how sharply the budget piles onto the worst clusters. (iv) **Fill each quota**. A funded cluster draws images by cosine similarity between its centroid and a grey-masked cutout of each candidate foreground. Appendix A gives the rest.

3.3 CONDITIONS

Four *arms* — training conditions — share one base pool and differ only in what is appended. **A0**, the unpadded baseline, is the base pool of 4447 pairs alone — an *unclean* control, differing in pool size as well as content, and *not* the configuration corresponding to the published row, which is C10 (Section 5). **A2** adds the 1000 original photographs of arm B’s stems, separating pool size from generated content. **B** adds 1000 renders drawn uniformly at random, redrawn per seed; **C10** adds the 1000 chosen by the allocation above. A2, B and C10 all hold 5447 pairs, and the claim this paper turns on is the gap between C10 and B. Appendix A records what was held constant.

4 SETUP, AND THE RULES WE FIXED IN ADVANCE

Every decision rule below was committed at a named commit before the runs it governs; amendments are append-only and dated; thresholds that failed are reported failed, and two that *passed* are reported and then refused.

Endpoints and seeds. The primary endpoint is COD10K-test S_α from the final round’s best teacher checkpoint; NC4K S_α is *reported, never decides*, as are MAE, F_β^w and E_ϕ . Seeds are $\{42, 43, 45\}$. CHAMELEON is not an endpoint, for a reason that is itself a result of this paper (Section 7).

The decision rule. Three runs of the same arm do not return the same score, and the size of that seed-to-seed wobble is what any real difference has to beat. Let $\hat{\sigma}$ be that wobble — the pooled within-arm standard deviation of S_α across arms, at 8 degrees of freedom:

REAL EFFECT iff $\Delta > 2\hat{\sigma}$ and sign consistent 3/3 seeds
 WITHIN NOISE iff $|\Delta| \leq 2\hat{\sigma}$
 REAL REGRESSION iff $\Delta < -2\hat{\sigma}$ and sign consistent 3/3 seeds
 INCONCLUSIVE iff $|\Delta| > 2\hat{\sigma}$ but sign not 3/3 — report as is, do *not* add seeds

So WITHIN NOISE means a gap smaller than the seed-to-seed spread, not a gap of zero. The rule is applied independently per architecture — SINet primary, SINet-v2 the robustness check, neither overriding the other, a disagreement reported as a disagreement. At $n = 3$ we report no p -value, bootstrap or multiple-comparison correction: verdicts are Δ against a $2\hat{\sigma}$ bar with sign-consistency counts, and nothing converts them into significance claims.

Two further arms that isolate direction from shape. Arms C10 and B differ by *concentration* as well as by targeting, so that comparison alone cannot speak to the uncertainty signal. A separate campaign of 12 runs — two architectures $\times$ two new arms $\times$ three seeds, B and C10 re-scored rather than re-trained — holds the allocation’s shape fixed and varies only direction. **CSHUF** permutes the per-cluster scores across clusters, destroying targeting; **CINV** reverses their rank, making the allocation maximally anti-targeted. CSHUF’s permutation is the first of 64 candidate draws with zero fixed points and $|\rho| \leq 0.10$; its realised rank correlation with the true score vector is -0.03351 , against -1.0 for CINV. Because equality is the *supporting* outcome here, set distinctness was gated

Table 1: The verdict on COD10K-test S_α , three seeds per arm. $\hat{\sigma}$ is pooled within-arm at $df = 8$; WITHIN NOISE means $|\Delta| \leq 2\hat{\sigma}$. The architectures disagree on the sign of the decisive gap. Per-run values: Table 6. *Source:* EXP ABC #3 @ REBUILD.LOG.txt:1376.

	$\hat{\sigma}$ ($df = 8$)	$2\hat{\sigma}$	$\Delta(B - A2)$	$\Delta(C - B)$	Verdict
SINet	0.0090	0.0179	+0.0058 3/3	+ 0.0050 3/3	WITHIN NOISE
SINet-v2	0.0061	0.0123	+0.0030 3/3	- 0.0004 2/3	WITHIN NOISE

before any GPU time at a Jaccard bound of 0.50: between 452 and 511 of each pair’s 1000 injected images differ.

Provenance of the rules. Appendix A gives the commit carrying each rule and all three amendments, including one clause that no correct computation could satisfy.

5 THE NULL RESULT

Targeting did not beat random. Under the rule of Section 4, applied to 24 training runs, $\Delta(C - B)$ is WITHIN NOISE on both architectures and both endpoints (Table 1), and the secondary endpoint returns the same verdict on every gap.

The honest limit: this campaign resolved coarsely. A verdict of WITHIN NOISE is only as strong as the noise, and here the noise was large enough that the bar a gap had to clear was *wider* than the effect the campaign was built to detect: $2\hat{\sigma} = 0.0179$ against this repository’s reproduction gap of 0.0142, which it was declared able to half-resolve. At 126% of that effect, **this campaign could not have detected the reference effect had it been present**, and the null survives only on the tighter bar imported from the falsification campaign.

That bar is $3.24\times$ tighter, $\hat{\sigma} = 0.0028$ against 0.0090, pooled within-arm over $\{B, C10, CSHUF, CINV\}$ at $df = 8$; it applies here because that pool contains B and C10. $\Delta(C10 - B) = 0.0050$ **against a bar of 0.0055 is $0.91\times$ threshold, 3/3 sign-consistent: still WITHIN NOISE, and marginally so.** The asymmetry matters: where equality is the *supporting* outcome a tighter bar is conservative, but here the claim and the null coincide, so the looser bar is anti-conservative. Even the tighter one resolves only 39% of the reference gap on SINet and 45% on SINet-v2.

The instrument is not blind. A null is only worth reading if the same measurement can detect something real, so we checked that it can: we trained an arm we expected to be clearly worse and asked whether this design would see the difference. It did. A mean-teacher arm — SINet, three seeds, base pool, single-round by construction — reaches 0.6988 on COD10K-test against C10’s 0.7185: $\Delta = +0.0197$ against a bar of 0.0060, or $3.3\times$ the bar, sign-consistent 3/3, verdict DETECTED, the secondary endpoint agreeing.

This is a control on *instrument sensitivity*, not an ablation — the arm differs from C10 in method, round count and pool size at once, so no component attribution follows and none is made. **And it does not close the gap that matters most.** It shows effects of roughly the reference magnitude are visible; it does *not* show an effect the size of our nulls would be. The control’s own bar, 0.0060, exceeds the decisive gap $\Delta(C10 - B) = 0.0050$, so an effect that size would not have cleared this pooling either. A positive control at 0.020 licenses *the instrument works, not it would have seen the effect we are nulling*.

Two loose ends, neither of which flatters us. Both of the campaign’s largest effects sit just inside the bar: the decisive gap is $0.91\times$ threshold, and the largest falsification gap, 0.0050 on *NC4K*, is also $0.91\times$ its own bar (on the primary endpoint the maxima are smaller, 0.0027 and 0.0024). Two WITHIN NOISE verdicts at 91% of threshold is not a comfortable null; a little more sensitivity would flip one to INCONCLUSIVE. And the ordering $A0 < A2 < B \approx C10$ is monotone in all four cells, three steps sign-consistent 3/3: if real, padding the pool helps and targeting adds nothing. No step clears its bar, but we report it because it points away from our conclusion.

Where these arms sit against the published numbers. Our C10 measures $S_\alpha = 0.7185$ against the 0.7136 reported for the corresponding published configuration. A0 is the *unpadded* baseline, not that configuration, and it passed its reproduction gate; Appendix H gives that gate, A0’s spread, and an untested pool-size mechanism for its instability.

6 WHAT GENERALIZES

The null does not establish that the approach fails, only that no benefit is resolvable at this sensitivity; it says nothing about the uncertainty signal, since C10 and B differ by concentration as well as targeting; and its scope is inherited from the generator. Those three gaps are what this section measures. Some of what follows is a quirk of our setup and some travels; Table 7 in Appendix I assigns every cause.

6.1 CONCENTRATING THE DATA MIMICS TARGETING

Suppose you aim a generation budget at the clusters where the model is least confident. The training set you get really does look different from a randomly chosen one of the same size: train a probe to tell the two apart and it succeeds easily, on images it has never seen. It is tempting to read that as evidence that the targeting worked. It is not.

We re-ran the selection with the uncertainty scores shuffled between clusters, so the budget was spread exactly as unevenly but pointed at the wrong places. The separation survived. We aimed at an arbitrary cluster instead: it survived again. Only removing the unevenness itself, by drawing two random sets of equal size, collapsed it (Figure 1a). The probe was detecting concentration, not the uncertainty signal.

The numbers follow the picture. Targeted selection separates from a random set of equal size at a held-out Cohen’s d — how far apart two sets are in standard deviations, scored on data the probe did not see — of roughly 1.00 to 1.23, an order of magnitude above a previously claimed 0.10. Permuting scores across clusters gives +0.9139 to +1.1403 in the same cells, and an *arbitrary* cluster as much again. Paired across twenty cells the signal adds +0.0073 of a d against its own shuffle, winning 13/20 ($P \approx 0.13$ under a fair coin) — weak evidence for a small positive contribution, not the coin flip we previously called it — and is *negative* at -0.0649 against an arbitrary cluster, winning 4/20 ($P \approx 0.006$). On the same yardstick the strongest directional evidence points *against* targeting. What targeting buys is narrowness — an effective-rank ratio of 0.634 over 20/20 cells, 0.53 to 0.64 in the four peak cells — covering no more of the target manifold.

Geometry is not performance, so we trained it too. The falsification arms hold concentration *exactly* fixed — CSHUF permutes the per-cluster scores and CINV reverses their rank, both reproducing the committed allocation’s shape to five decimal places on all four recorded shape statistics — and vary only direction. All twelve cells are WITHIN NOISE, the largest gap anywhere being 0.0050 at $1.82\hat{\sigma}$. A directional pattern exists and we report it without claiming it: C10 is best in none of the four cells, the anti-targeted CINV has the highest mean in three of four, and the architectures disagree in sign on two of three gaps — the signature of noise. None of it reaches the bar, so it is not the verdict.

Scope, and the trained half is the weaker one. Negligible *in this geometry* is supported; zero and harmful are not. The trained comparison is attenuated three times over, and the factors multiply: injected images are 1000 of a 5447-image pool (18.4%), the pinned budget dilutes exposure, and only $\approx 48\%$ of injected images differ between arms — an effective contrast near 8.8% against a bar of 0.0055. **So the 12/12 trained result is weakly informative, and the geometric result carries this claim.** That attenuation biases toward the null, which is why we disclose it. **Transfers:** the control that matters is not random selection but a same-shape allocation with the signal destroyed.

6.2 THERE WERE NO CLUSTERS TO AIM AT

Allocating cluster by cluster presumes the data falls into clusters. It barely does. The silhouette score — a standard measure of how cleanly points fall into groups, where values near zero mean there are no real groups — peaks at 0.1600 at $k = 75$ in the space the pipeline uses, and the other two spaces we tried reach only 0.1465 and 0.0568, all below any conventional threshold, so the weakness is not an artifact of one encoder. The third fails more specifically: its silhouette falls

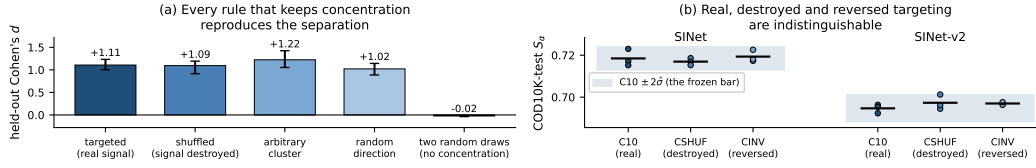


Figure 1: **(a)** How far a selected set sits from a random set of the same size, in held-out Cohen’s d . Every rule that keeps the allocation’s *concentration* reproduces the separation — permuting the uncertainty scores across clusters, and ignoring them entirely — and only destroying concentration collapses it. Bars are means over four cells, whiskers the range. **(b)** Trained S_α for the three arms that hold concentration *exactly* fixed and vary only the signal’s direction; points are seeds, rules are arm means, the band is the $\pm 2\sigma$ bar. *Source:* `c1_attribution.csv` (EXP C1 #4); `out/t2/abc_metrics.csv` (EXP T2 #3).

monotonically from 0.0568 at $k = 5$ to 0.0357 at $k = 150$, returning its smallest grid value by hitting the boundary rather than finding a maximum. Our guard required an interior maximum, and it failed. Two measurements reinforce this: an arbitrary preprocessing choice reassigns 5.4% of memberships, so an effect smaller than that cannot be attributed to targeting; and the pool is the two-dataset mixture of Section 3.1, separating from itself at up to 0.9225. **Transfers:** the check costs one silhouette sweep and should precede any method that allocates over clusters.

6.3 UNCERTAINTY TRACKS PIXEL ERROR BETTER THAN SHAPE

An uncertainty score is only useful if it points at the errors you care about. In camouflage that means the boundary, which is what S_α rewards and what a per-pixel average like MAE largely ignores. These signals track the per-pixel average better.

The ordering $\rho(\text{MAE}) > \rho(1 - S_\alpha) > \rho(1 - \text{IoU})$ holds in **six independent signal–architecture tests**: three estimators — student–teacher disagreement, predictive entropy and ensemble disagreement — crossed with two architectures. The table carries eight rows because the ensemble is evaluated on two arms, but our own protocol treats the second as a *sensitivity check* on the first, so they are not independent and we do not count them twice. The gap spans +0.0708 to +0.3330, all clearing the resolution floor of about 0.06, the smallest only just. The same criterion can fail, and elsewhere it does: aggregated over a boundary band, only 2 of 8 rows pass and four go negative, outside our declared interpretation space, which did not anticipate inversion.

The area control. Our own explanation for the boundary collapse — a whole-image mean is uncertainty density times area — *applies to the whole-image rows too*: if MAE and the signals both scale with area while S_α carries an object-size term that does not, the ordering could be an artifact of the metrics. So we partialled area out, recomputing each correlation with size held constant (Table 5).

The evidence against the confound is the direction of the shift, not the pass count: partialling object area raises $\rho(1 - S_\alpha)$ in all eight rows, the opposite of what the confound predicts — area was suppressing the structural correlation, not inflating the pixel one. The ordering then survives on 7/8 and 6/8 rows against a floor of 6/8 declared before the run. That floor is weak and we say so: 6 of 8 has $P \approx 0.145$ under a fair coin, so the uncertain-area covariate clears a near-chance bar by zero margin. **Area accounts for part of the gap but not its sign.**

Scope: correlational, one endpoint, and the *magnitude* does not generalise — $\rho(1 - S_\alpha)$ reaches +0.5628 on SINet-v2, so the claim is strictly ordinal. **Transfers:** if you care about structural quality, uncertainty of these kinds is a better proxy for pixel-average error than for what you want to improve.

6.4 REASONS SPECIFIC TO OUR SETUP

The reasons above would apply to anyone. Three more are ours alone, properties of the particular loop and generator we used. The optimisation budget is pinned at 253 and 127 steps per epoch

Table 2: Which evaluation columns can be reported as independent measurements. Every rate is a *lower bound*: only same-dimension images are comparable, so an image with no candidate is unchecked, not clean; crops, rescales and different photographs of one specimen are not detected. In the exact-hash column 0 is a measured zero, “—” not applicable. *Source*: CLEAN_PROTOCOL.md; EXP D2 #5, D2R #4, D2_NC4K #2.

Endpoint	Contaminated	Exact hash	Unchecked	Nearest-neighbour gap	Verdict
COD10K-test	2/2026 (0.1%)	—	524/2026	none	reportable [†]
NC4K	1/4121 (0.0%)	—	2406/4121	none	reportable [†]
CHAMELEON	41/76 (53.9%)	0	25/76	41 below 5.51, next 40.58	not reportable
CAMO	4/250 (1.6%)	—	155/250	none	selection set only

[†] Measured against on-disk copies, not author-sourced ones — the same weakness the CHAMELEON re-audit closes. That re-audit is *owed* for these rows; until then their rates are provisional.

regardless of pool size, so adding 1000 images bought no extra optimisation, only a change of mixture. The foreground supply is exhausted: the render set is a bijection onto the base set and not one traced object shows any sign of regeneration, so the objects are copied rather than synthesised. And the generated distribution is narrower than its own input — real photographs of a different genre cover the target manifold better than synthetic images generated from those very photographs.

Each has an obvious rebuttal and each rebuttal is right: the first is a property of the training schedule, not of synthetic data; the second follows from choosing a generator that composites by design; the third is a distributional characterisation, so “the distribution is far, therefore no amount of it can help” would be a non-sequitur. None licenses the conclusion that closed-loop generation cannot work; they say only that this build could not have shown that it does. We also *withdraw* one of our own candidate causes, by source reading rather than measurement. Appendix C gives every measurement, the withdrawn cause, and the evidence that the coverage result alone does not depend on statistical power.

7 BENCHMARK CONTAMINATION AND A CLEAN PROTOCOL

CHAMELEON is one of the four benchmarks camouflage papers report on, and more than half of it is training data. The usual check hashes the image files and looks for identical bytes; run it here and it reports no overlap at all. It is not wrong — it answers a different question. The copies were re-saved in a different format, so not one byte matches while every pixel does. Nothing here depends on whether Section 3’s method worked.

The finding, and the control that misses it. 41 of CHAMELEON’s 76 images — 53.9% — are the same photographs as images in the COD10K-train and CAMO training pool, re-encoded, 40 in the public COD10K-train split and one in CAMO. A model trained on COD10K-train has already seen over half of CHAMELEON before being evaluated on it. We verified against an author-sourced release, byte-identical at 76/76. Exact hashing returns 0 collisions and is correct at that level: these are re-encodings, not byte-identical copies. The number was right; the inference drawn from it was not — and we hit the same trap ourselves, our first implementation reporting 10/76 (13.2%) because contrast normalisation destroyed the discriminative scale. CHAMELEON predates COD10K, so the parsimonious reading is that COD10K-train absorbed these images, though only pool membership is measured.

The count is a property of the data, not of a tolerance. Matches at mean absolute difference $\leq 1, 2, 3, 5, 6$ are 11, 26, 37, 40 and 41 of 76: the count saturates. The sorted nearest-neighbour distances independently place 41 images below 5.51 with the next at 40.58 — a $7.36\times$ jump — unchanged at shortlist depth 8, 32, or all. Two instruments that owe each other nothing land on the same integer, and the other three endpoints show no jump.

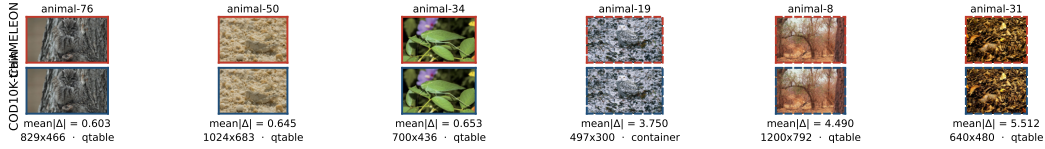


Figure 2: Six of the 41 confirmed pairs: the author-sourced CHAMELEON image (top) and its partner in the COD10K-train pool (bottom). *Selection rule, applied not curated:* the six lowest mean absolute difference among pairs with a COD10K-train partner. Exact hashing reports 0 collisions for every pair shown, each being a re-encoding rather than a byte-identical copy. *Source:* chameleon_contaminated.json (EXP D2R #4).

A negative control, from the same instrument. An audit that finds contamination wherever it looks is worthless. The identical detector — imported, not re-implemented — returns a clean null on NC4K against the COD10K test split: 0 byte-identical, 0 pixel-identical, 0/4121 re-encoded at every tolerance swept (0 of the 1082 images actually checkable against it), and a minimum nearest distance of 20.610, 3.44 $\times$ the confirmation tolerance. NC4K’s 1/4121 in Table 2 is a *different axis*, the full training pool, against which 1715 of its images are checkable; the two must not be collapsed.

Non-independence, not inflation. We expected to say how much a contaminated column flattens a score, measured it, and found nothing: all eight difficulty percentiles for the leaked subset land within 0.448 to 0.559, and the sign of the clean-minus-leaked difference *flips* with the mask release. The pre-declared reading — a percentile inside [0.25, 0.75] means no skew — is what stopped us writing the quotable sentence. Our claim is therefore non-independence, not inflation: *CHAMELEON is not an independent endpoint for COD10K-trained models.*

The protocol: retire the benchmark, do not subset it. Our released protocol originally recommended reporting CHAMELEON on its 35 uncontaminated images. We withdraw that: of 76 images, 41 are contaminated and 25 more have no same-dimension candidate and are *unchecked, not clean*, leaving 10 **verifiably clean**. A 10-image endpoint is not an endpoint, and with two incompatible mask releases the honest recommendation — a change to one we already released — is to retire CHAMELEON rather than subset it. Report COD10K-test and NC4K instead, disclosing the unchecked share. The detector we release is checkable without our data, and Appendix F records its self-test. [TODO: author finalize — anonymised release URL for the detector, the contaminated-pair list, and the clean protocol.]

A concrete instance, from the protocol we test. The method under test evaluates on CHAMELEON, and its task setup states that for this family of benchmarks it “*consistently use[s] the real unlabeled COD10K training set as the target domain*”. It takes a precaution: when the test set is CHAMELEON, those images are removed from the *source* domain “*to prevent target domain data leakage*”. That closes the source path but not the target pool, since the case definitions adjust source selection only — and that pool holds 40 of the 76 CHAMELEON images, so a model following this protocol self-trains on pseudo-labels over a set containing more than half the images it is then tested on. We attribute no intent: the source-side removal shows the authors were alert to leakage, and the target-side path is invisible without a re-encode-tolerant check.

8 CONCLUSION

We built a loop that aims a synthetic-data budget by the model’s own uncertainty, and found no gain we could resolve — at 39%–45% of the improvement reported for this approach, with a positive control confirming the design detects a real effect of +0.0197 but not one the size of the null. The idea still seems reasonable; for it to have a chance you would need an optimisation budget that grows with the pool, a foreground supply not already exhausted, and a generator wider than its own input. Separately: CHAMELEON is not an independent test set for models trained on COD10K, the usual check will not tell you so, and we release one that will. A control that returns 0 is only as good as its ability to return something else.

AI USE STATEMENT

[TODO: author finalize — this statement is required by ICLR 2027 and must describe the authors’ actual use of generative AI tools. It must be written by the authors, not inferred, and it should state which tasks with required disclosure such tools were and were not used for, which are not applicable, and how AI-assisted work was verified. One verification mechanism is available to this work and should be named if relevant: every numeric value in this paper is traceable to a committed log block emitted by a committed script, and no result was transcribed by hand; the traceability map in Appendix B records the mapping. The authors take responsibility for the final content of this work.]

ETHICS STATEMENT

This work involves no human subjects and no personally identifying data. It uses public research benchmarks for camouflaged object detection under their published terms [TODO: author confirm — dataset licence statements for each benchmark used].

Section 7 reports an overlap between two public datasets. We state explicitly what that measurement does and does not support. It is a measurement of pool membership; it attributes no intent to any dataset’s authors, and we draw the direction of provenance as an inference from release chronology rather than as a finding. We measured for a score-inflation figure, did not find one, and do not report one. The one correction we direct at anyone is directed at ourselves: it was our own protocol that adopted CHAMELEON as an evaluation endpoint without checking whether the repository we built on ever sanctioned it. We release the detector so that the measurement can be repeated, contested, or extended by others rather than taken on trust.

REPRODUCIBILITY STATEMENT

The decision rules governing every result are committed before the runs they govern, at the commits named in Section 4, and amendments to them are append-only and dated. Each experiment has a committed result document and one or more committed log blocks; each number in this paper is traceable to such a block or to a committed artifact, and Appendix B gives that mapping. Per-run metrics for all 36 training runs, the full clustering sweep, and the full signal-generalizability table are in the appendix. The contamination detector is released with an eight-assertion known-answer self-test that runs without any of our data [TODO: author finalize — anonymised URL for code, artifacts, and the released detector].

Two limits on reproducibility. Our regeneration check establishes byte-identical reproduction on one machine with one software stack; it is not a claim of determinism across hardware or driver versions. And a render from this generator is a function of its foreground, its mask, *and its position within the processing shard*, so changing the shard count changes the outputs even at a fixed seed.

A METHOD SPECIFICATION

Training. SInet at batch 16 for 40 epochs and SInet-v2 at batch 32 for 100 epochs, Adam at 10^{-4} , images at 352×352 with ImageNet normalisation. The EMA teacher uses momentum 0.996 under this task setting. The consistency weights in Equation 1 are the configured $a = 0.9$, $b = 0.3$ with the unweighted cross-entropy variant — note that these are the values the trainer sets for this task, not the loss class’s own defaults. Strong views add random autocontrast and Gaussian blur to the weak view. `cuda.deterministic` is enabled and `cuda.benchmark` disabled; this reduces, and is not claimed to eliminate, kernel nondeterminism.

What was held constant, verified. Across all 24 runs the base pool is asserted byte-identical against a hash manifest with 0 mismatches, both rounds read the unfiltered 4040-image target set, and every run shares one trainer invocation, checkpoint rule, endpoint set and deterministic-kernel setting.

Optimisation budget. Steps per epoch are set by the shorter of the source and target loaders, which fixes them at 253 (SINet) and 127 (SINet-v2) in both rounds of every run regardless of pool size. This is measured per run and asserted, and it is the mechanism behind Section 6.4’s first cause.

Allocation sweeps. Stage (iv) matches against a grey-masked cutout rather than a render because at selection time the foreground exists and the render does not. The budget grid is $B \in \{250, 500, 1000, 2000, 3000, 4447\}$ and the temperature grid over α is extended until both degenerate ends are reached, defined numerically as concentrated when $\max_j p_j \geq 0.95$ and uniform when the total variation from uniform falls below 0.01, with caps at $\alpha \in [10^{-4}, 10^3]$. The campaign uses $\alpha = 1.0$, $B = 1000$.

Metrics. S_α at $\alpha = 0.5$ is the primary endpoint metric; MAE is the checkpoint-selection metric; F_β^w and E_ϕ are computed and reported but decide nothing. The F -measures apply β in the non-squared form, so the metric name rather than a formula is what identifies them.

The A2 control’s one confound. A2 pairs the authors’ original photographs with the authors’ masks, whose mean foreground fraction differs from the raw masks by 0.0058 on 1000 of 5447 images. This affects $\Delta(B - A2)$ and is disclosed; it does not touch $\Delta(C10 - B)$, the decisive gap, since B and C10 share mask provenance, pool size and render pool.

Pre-registration commits and amendments. The four-arm rule is recorded at 065dac6, the commit carrying that campaign’s first log block; the falsification campaign’s at 617a2e1, before its first training run; the signal-generality check’s at 6efde9f; the area control’s at c2114af; the positive control’s at c2114af. Three amendments exist, all append-only and dated. Two are reference-tolerance defects of the same kind: a frozen clause requiring reproduction to better than 10^{-9} against a file recording six decimal places, and — in the area control, written by the same author who had just documented the first — the identical mistake against a file recording four. Both are replaced by exact agreement at the recorded precision, which is the strictest satisfiable form, and neither altered a decision threshold. The third marks one verification gate not applicable in scope after it was found to compare two architectures’ signals.

B SUPPORTING TABLES

The area control in detail. The covariate is object area on the endpoint side and uncertain-region area on the target side. Partialling object area raises $\rho(1 - S_\alpha)$ in all eight rows — +0.3433 to +0.4988 for student–teacher disagreement on SINet. The single object-area failure is a tie crossing by 0.0012, and both uncertain-area failures are on SINet-v2, where $\rho(\text{MAE})$ falls +0.7070 to +0.4457 once band area is removed.

C CAUSES SPECIFIC TO OUR SETUP, IN FULL

These causes are properties of this loop and this generator. They explain why this attempt could not have demonstrated a gain; they do not license the conclusion that closed-loop generation cannot work. Each carries a legitimate dismissal, which we state.

The optimisation budget is pinned at 253 and 127 steps per epoch for the two architectures, in both rounds of every run, regardless of pool size — so adding 1000 images to a 4447-image pool bought no additional optimisation, only a change of mixture. A reviewer may call this a property of the training schedule rather than a result about synthetic data, and that reading is correct: nothing here measures what happens when the optimisation budget grows with the data, which is the largest open question this paper creates.

The foreground pool is exhausted; the background is not. The render set is a bijection onto the base foreground set — all 4447 renders trace into it, 0 outside, 0 base foregrounds unrendered — on the pool training reads as well as on ours, and the mapping is literal: interior error against the source foreground is 5.603 and 9.633 mean absolute grey levels against background error near 70.6. Of the 8885 objects with a valid interior comparison (4443 + 4442, since erosion leaves no interior at all for four and five objects each under 1% of frame), 0 show any sign of regeneration. This is

Table 3: Full clustering sweep over the target set: silhouette (mean over 10 k -means seeds), bootstrap ARI and seed ARI, for every k in the declared grid and all three embedding spaces. The principled k per space is **bold**. No k in any space reaches a silhouette conventionally regarded as well-separated structure, and clipL224 falls monotonically, so its selection criterion returns the grid edge rather than an interior maximum — recorded as a FAIL of the declared guard. *Source:* rebuild/B1/out/b1_k_sweep_{dinoL224,dinoL518,clipL224}.csv; block EXP B1 #3 @ results/REBUILD-LOG.txt:922.

k	dinoL224			dinoL518			clipL224		
	sil.	bs.ARI	sd.ARI	sil.	bs.ARI	sd.ARI	sil.	bs.ARI	sd.ARI
5	0.0538	0.712	0.795	0.0566	0.624	0.732	0.0568	0.953	0.983
10	0.0868	0.676	0.709	0.0923	0.598	0.770	0.0550	0.692	0.805
15	0.1053	0.608	0.714	0.1152	0.573	0.592	0.0486	0.575	0.670
20	0.1180	0.629	0.693	0.1326	0.506	0.524	0.0486	0.635	0.646
30	0.1378	0.596	0.642	0.1527	0.539	0.617	0.0488	0.609	0.631
50	0.1465	0.610	0.615	0.1556	0.579	0.600	0.0448	0.498	0.549
75	0.1457	0.588	0.627	0.1600	0.579	0.616	0.0396	0.463	0.477
100	0.1419	0.608	0.637	0.1557	0.583	0.601	0.0382	0.419	0.452
150	0.1367	0.630	0.636	0.1507	0.618	0.625	0.0357	0.380	0.409

Table 4: Full signal-generality measurement, all 16 rows: four signal variants across three signal families, two architectures, two aggregations. ρ at the committed seed, $\pm$ sd over 10 k -means seeds. The whole-image block passes the pre-registered ordering 8/8; the boundary block passes 2/8 with four rows going negative, an outcome outside the pre-registered interpretation space, reported as it fell. *Source:* rebuild/T2C/out/t2c-table.csv; block EXP T2C @ results/REBUILD-LOG.txt:2550.

arch	signal	agg.	$\rho(\text{MAE})$	$\rho(1 - S_\alpha)$	$\rho(1 - \text{IoU})$	order	seeds
SINet-v2	ES	whole	+0.7070 $\pm$ 0.0404	+0.5628 $\pm$ 0.0359	+0.4591 $\pm$ 0.0315	PASS	10/10
SINet-v2	ensemble A0	whole	+0.6191 $\pm$ 0.0675	+0.5483 $\pm$ 0.0409	+0.4540 $\pm$ 0.0451	PASS	9/10
SINet-v2	ensemble CSHUF	whole	+0.5576 $\pm$ 0.0690	+0.4488 $\pm$ 0.0580	+0.3370 $\pm$ 0.0577	PASS	10/10
SINet-v2	entropy	whole	+0.7393 $\pm$ 0.0309	+0.5404 $\pm$ 0.0408	+0.4070 $\pm$ 0.0395	PASS	10/10
SINet-v2	ES	bound	−0.5485 $\pm$ 0.0745	−0.1146 $\pm$ 0.0675	+0.0537 $\pm$ 0.0558	FAIL	0/10
SINet-v2	ensemble A0	bound	−0.1960 $\pm$ 0.0524	+0.0425 $\pm$ 0.0410	+0.0933 $\pm$ 0.0396	FAIL	0/10
SINet-v2	ensemble CSHUF	bound	−0.2527 $\pm$ 0.0668	−0.0591 $\pm$ 0.0609	−0.0011 $\pm$ 0.0579	FAIL	0/10
SINet-v2	entropy	bound	+0.4208 $\pm$ 0.0356	+0.4182 $\pm$ 0.0455	+0.3605 $\pm$ 0.0470	PASS	8/10
SINet	ES	whole	+0.6284 $\pm$ 0.0381	+0.3433 $\pm$ 0.0301	+0.2613 $\pm$ 0.0383	PASS	10/10
SINet	ensemble A0	whole	+0.6028 $\pm$ 0.0466	+0.2698 $\pm$ 0.0487	+0.1491 $\pm$ 0.0560	PASS	10/10
SINet	ensemble CSHUF	whole	+0.5212 $\pm$ 0.0564	+0.4006 $\pm$ 0.0459	+0.3361 $\pm$ 0.0447	PASS	10/10
SINet	entropy	whole	+0.5892 $\pm$ 0.0368	+0.4267 $\pm$ 0.0277	+0.3588 $\pm$ 0.0351	PASS	10/10
SINet	ES	bound	−0.0012 $\pm$ 0.0481	−0.0567 $\pm$ 0.0500	−0.0554 $\pm$ 0.0479	FAIL	8/10
SINet	ensemble A0	bound	+0.3633 $\pm$ 0.0605	+0.2910 $\pm$ 0.0509	+0.2215 $\pm$ 0.0513	PASS	10/10
SINet	ensemble CSHUF	bound	+0.3175 $\pm$ 0.0637	+0.3857 $\pm$ 0.0381	+0.3770 $\pm$ 0.0322	FAIL	2/10
SINet	entropy	bound	+0.2434 $\pm$ 0.0414	+0.2492 $\pm$ 0.0540	+0.2432 $\pm$ 0.0563	FAIL	6/10

an identity statement about pixels, not a claim about similarity, which is what makes it resistant to the obvious counter-argument. The dismissal — this generator composites by design, so of course it adds no new foregrounds — is right, and what survives it is the scope statement every null here inherits: evidence about targeting under an exhausted foreground supply, silent on whether a larger or more diverse one would behave differently.

The generated distribution is narrow, and this evidence alone does not depend on power. As recall against the target manifold the synthetic pools reach 0.1277 to 0.5421 across three embedding spaces, while the generator’s own input pool — real photographs of a different genre — reaches 0.7097 to 0.7473 and a different real camouflage dataset reaches 0.7921 to 0.8995: real photographs of the wrong genre cover the target better than synthetic images generated from those very pho-

Table 5: The area control (EXP AC). First-order partial Spearman on the same cluster partition and floors as the committed measurement. z_{obj} = per-cluster mean object area on the endpoint side; z_{unc} = per-cluster mean uncertain-region area on the target side. Ordering survives 7/8 and 6/8 against a floor of 6/8 declared before the run. Note that partialling object area *raises* $\rho(1 - S_\alpha)$ in every row — the opposite of the confound’s prediction. *Source*: rebuild/AC/out/ac.table.csv; block EXP AC @ results/REBUILD-LOG.txt:2612.

arch	signal	raw		$ z_{obj} $		$ z_{unc} $	
		MAE	$1 - S_\alpha$	MAE	$1 - S_\alpha$	MAE	$1 - S_\alpha$
SINet	ES	+0.6284	+0.3433	+0.5761	+0.4988	+0.5518	+0.3897
SINet	entropy	+0.5892	+0.4267	+0.5505	+0.5517	+0.5060	+0.4426
SINet	ensemble A0	+0.6028	+0.2698	+0.5411	+0.4301	+0.5603	+0.2583
SINet	ensemble CSHUF	+0.5212	+0.4006	+0.5002	+0.4857	+0.4857	+0.3937
SINet-v2	ES	+0.7070	+0.5628	+0.7062	+0.6272	+0.4457	+0.4744
SINet-v2	entropy	+0.7393	+0.5404	+0.7134	+0.6546	+0.5369	+0.4266
SINet-v2	ensemble A0	+0.6191	+0.5483	+0.6245	+0.5969	+0.4355	+0.4488
SINet-v2	ensemble CSHUF	+0.5576	+0.4488	+0.5344	+0.5209	+0.4482	+0.3615
ordering holds		8/8		7/8		6/8	

tographs, a paired loss of -23.6% to -82.9% . Effective rank falls 243.3 to 193.2 while precision rises 0.6490 to 0.6886 — narrowing, not displacement. This is a distributional characterisation, not a training-utility bound; reading it as “the distribution is far, therefore no amount of it can help” would be a non-sequitur, and what generalises is the design, not the number.

One candidate cause we removed rather than kept. A convenient explanation was available: that foreground information reaches the regenerated background only through a narrow low-dimensional summary. Examining the released generator’s architecture refutes that account as stated — four routes carry foreground information into the regenerated region, the narrow summary is one, and the dominant route bypasses it entirely. *This is resolved by source reading, not by measurement*: no experiment was run and no metric logged, and it withdraws one of our own candidate causes rather than establishing a result. Showing the channel is not narrow does not show its capacity is used, so we decline both the claim that the generator is not the binding constraint and that it is steerable.

D CONTAMINATION: CORRECTIONS TO OUR OWN WORK

Corrections to our own work. An author-sourced re-audit reproduced 8 of 8 of our earlier measurements; its three mismatches correct our prior work, not the data. One declared threshold is left *failing* rather than reworded: we required every confirmed pair to differ in its JPEG quantization table and compared with an inequality that scored a *missing* table as a differing one — absence of evidence read as evidence — because two files are PNGs carrying a .jpg extension. Corrected, 41 of 41 pairs still carry independent re-encoding evidence, 40 by quantization table and one by container format. Separately, CHAMELEON was never a sanctioned endpoint in the repository we built on; it was our own plan that promoted it, so the framing here is not that we found a defect in someone else’s evaluation.

E THE ENDPOINT-MEASURED SIGNAL

A correction to our own measurement. Our first measurement correlated this signal *on the labelled endpoint* against endpoint error — not the quantity the pipeline allocates by. Measured on the signal actually available, correlation with pixel error falls by +0.2100 and +0.2470, from +0.8695 and +0.8754 to +0.6595 and +0.6284. A signal validated where labels exist overstates what is usable where they do not.

F THE TWO CHAMELEON MASK RELEASES, AND THE RELEASED DETECTOR

The two circulating CHAMELEON ground-truth releases are different annotations, unrelated to contamination or to any method. They disagree in polarity, giving a mean IoU of 0.0 as stored and 0.6932 after polarity alignment; only 27 of 76 masks are exactly identical after alignment and 40 of 76 reach $\text{IoU} \geq 0.9$. On identical predictions the choice of release alone moves MAE by $2.7\times$ (0.2196 against the author-sourced masks, 0.0816 against the repackaged ground truth). Any CHAMELEON column is therefore not comparable across papers unless the release is named.

Detector self-test. The released detector passes an eight-assertion synthetic known-answer test using no repository data, reproduces the canonical 41 pairs and 41 names exactly, and surfaced one additional pair — the second partner of the one image that has two.

G A THRESHOLD THAT PASSED, AND WAS REFUSED

A threshold that passed, and was refused. On the secondary endpoint, $\Delta(A0 \rightarrow C10)$ for SINet is +0.0109 with sign consistent 3/3, against a bar of 0.0083: the frozen rule returns REAL EFFECT. We do not report it as one, on two independent grounds each sufficient by itself. NC4K is the secondary endpoint and decides nothing, and the comparison is against A0, which is both confounded in pool size and the least stable arm in the campaign. We record it prominently *because* it is the tempting one.

H ARM ANCHORING AND A0’S INSTABILITY

Where these arms sit against the published numbers. The published configuration corresponding to arm C10 reports $S_\alpha = 0.7136$ on this endpoint; our C10 measures 0.7185. A0 is the *unpadded* baseline and is not that configuration: its reproduction gate was on seed 42, which measured 0.7151 against this repository’s committed 0.7172, inside the declared $3\sigma = 0.0107$. The A0 arm *mean* of 0.7010 sits lower because A0 is the least stable arm, spanning 0.0334 across seeds on SINet and 0.0211 on SINet-v2. We offer a mechanism rather than leaving it unexplained: A0 holds 4447 images where every other arm holds 5447 under an identical pinned step count, so it samples a different mixture and should carry higher seed variance — and the effect reproduces on both architectures, which argues for a property rather than a broken run. We did not test this, and state it as a hypothesis.

I THE CAUSE SUMMARY

Table 6: Per-run metrics at full recorded precision for every training run in this work — the two frozen campaigns and the mean-teacher positive control — on both endpoints. Arms B and C10 appear once; they were re-scored, not re-trained, for the falsification campaign and reproduced their committed values exactly at six decimal places. *Source:* `rebuild/ABC/out/abc_metrics.csv` and `out/t2/abc_metrics.csv`; blocks EXP ABC #3 and EXP T2 #3.

arch	arm	seed	endpoint	S_α	MAE
SINet	A0	42	COD10K	0.715086	0.074960
SINet	A0	43	COD10K	0.681706	0.077044
SINet	A0	45	COD10K	0.706058	0.077347
SINet	A2	42	COD10K	0.705507	0.075994
SINet	A2	43	COD10K	0.708334	0.078489
SINet	A2	45	COD10K	0.709196	0.080858
SINet	B	42	COD10K	0.711554	0.076085
SINet	B	43	COD10K	0.713632	0.076278
SINet	B	45	COD10K	0.715155	0.075917
SINet	C10	42	COD10K	0.717165	0.073184
SINet	C10	43	COD10K	0.723034	0.075136
SINet	C10	45	COD10K	0.715243	0.078355
SINet	CINV	42	COD10K	0.722612	0.073910
SINet	CINV	43	COD10K	0.717343	0.077323
SINet	CINV	45	COD10K	0.718084	0.073607
SINet	CSHUF	42	COD10K	0.718602	0.074840
SINet	CSHUF	43	COD10K	0.716974	0.077678
SINet	CSHUF	45	COD10K	0.715291	0.079340
SINet	MT	42	COD10K	0.699633	0.093377
SINet	MT	43	COD10K	0.699167	0.087571
SINet	MT	45	COD10K	0.697503	0.091322
SINetv2	A0	42	COD10K	0.688813	0.091691
SINetv2	A0	43	COD10K	0.678740	0.095147
SINetv2	A0	45	COD10K	0.699883	0.085756
SINetv2	A2	42	COD10K	0.691914	0.089186
SINetv2	A2	43	COD10K	0.688977	0.094524
SINetv2	A2	45	COD10K	0.695414	0.087455
SINetv2	B	42	COD10K	0.698340	0.092242
SINetv2	B	43	COD10K	0.689481	0.089883
SINetv2	B	45	COD10K	0.697385	0.087132
SINetv2	C10	42	COD10K	0.692321	0.093395
SINetv2	C10	43	COD10K	0.696328	0.090749
SINetv2	C10	45	COD10K	0.695360	0.089182
SINetv2	CINV	42	COD10K	0.697008	0.088438
SINetv2	CINV	43	COD10K	0.696331	0.087293
SINetv2	CINV	45	COD10K	0.697605	0.090189
SINetv2	CSHUF	42	COD10K	0.701269	0.087506
SINetv2	CSHUF	43	COD10K	0.694490	0.087305
SINetv2	CSHUF	45	COD10K	0.696218	0.089485
SINet	A0	42	NC4K	0.765121	0.085752
SINet	A0	43	NC4K	0.751545	0.088874
SINet	A0	45	NC4K	0.757700	0.090228
SINet	A2	42	NC4K	0.760908	0.088385
SINet	A2	43	NC4K	0.762933	0.087466
SINet	A2	45	NC4K	0.765789	0.090892
SINet	B	42	NC4K	0.764108	0.087897
SINet	B	43	NC4K	0.768972	0.085001
SINet	B	45	NC4K	0.767632	0.088175
SINet	C10	42	NC4K	0.765296	0.086273
SINet	C10	43	NC4K	0.771241	0.085305
SINet	C10	45	NC4K	0.770503	0.086622
SINet	CINV	42	NC4K	0.774697	0.083144
SINet	CINV	43	NC4K	0.774114	0.084030
SINet	CINV	45	NC4K	0.768136	0.084570
SINet	CSHUF	42	NC4K	0.766713	0.085734
SINet	CSHUF	43	NC4K	0.767983	0.085825
SINet	CSHUF	45	NC4K	0.767229	0.089734
SINet	MT	42	NC4K	0.752869	0.108067
SINet	MT	43	NC4K	0.746708	0.101192
SINet	MT	45	NC4K	0.749400	0.104323
SINetv2	A0	42	NC4K	0.745691	0.092348
SINetv2	A0	43	NC4K	0.736067	0.099838

Table 7: Every cause, its measured quantity, and how far it carries. Causes marked *generalizes* concern the target data or the uncertainty signal and travel beyond this instantiation; those marked *specific to this setup* are properties of this loop or this generator. The scope column, not the magnitude column, determines how far a row may be carried. *Sources, by row:* FINAL_RESULTS.md Tables 8.2, 8.1, 9.2, 9.4, 11.1, 12.1, 5.1, 7.1 and 10.4; blocks EXP B1 #3 and #4, EXP C1 #4, EXP T2 #3, EXP T2C, EXP ABC #2, EXP D1, EXP A3, in results/REBUILD_LOG.txt. The conditioning row has no log block by construction.

Cause	Measured quantity	Scope	Carries to
No cluster structure to allocate over	best silhouette 0.1600; 0.1465, 0.0568 elsewhere	generalizes	any method allocating over clusters of such a distribution
Membership is not a robust label	5.4% of memberships move under a preprocessing change	generalizes	a floor beneath which an effect cannot be attributed to targeting
Signal adds nothing beyond concentration, in geometry	+0.0073 of a d against its own shuffle, 13/20	generalizes	any uncertainty-guided acquisition: the shuffle is the control
... and nothing resolvable on trained accuracy	12/12 WITHIN NOISE, 3.24 $\times$ tighter bar	generalizes	this signal class carries no resolvable directional information
Targeting buys proximity, not coverage	effective-rank ratio 0.53–0.64, 20/20	generalizes	concentration narrows the set rather than extending its reach
Pixel error predicted better than structural error	ordering passes 8/8 rows, 3 signals $\times$ 2 architectures	generalizes	uncertainty-guided allocation for this task in general
Endpoint-measured signal overstates itself	ρ drops +0.2100, +0.2470	generalizes	any signal validated where labels exist
Optimisation budget is pinned	253 / 127 steps, both rounds, every run	specific to this setup	nothing — a property of this loop
Foreground pool is exhausted	bijection 4447/4447; 0 outside, 0 unrendered	specific, but the scoping sentence travels	the scoping sentence any comparable null must carry
Objects are copied, not generated	interior 5.603 / 9.633 vs background 70.802 / 70.555	specific to this setup	the measurement design, reusable on any inpainting generator
Generated distribution is narrow	recall 0.13–0.54 vs 0.71–0.90 for every real set	specific to this setup	the design: a paired within-content coverage delta
Conditioning channel is <i>not</i> the bottleneck	four routes; source reading, no experiment	specific to this setup	nothing — a withdrawn cause